

TrES-3b

R-0367

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170605 · created 2026-07-18T15:07:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457909.8505 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.057
Transit depth (Rp/Rs)^2	3.52 +/- 2.13 [%]
Orbital Inclination (inc)	81.53 +/- 2.42
Airmass coefficient 1 (a1)	0.909 +/- 0.026
Airmass coefficient 2 (a2)	0.093 +/- 0.092
Scatter in the residuals of the lightcurve fit is	2.89 %
Best Comparison Star	#3 - [473, 325]
Optimal Method	PSF photometry
Transit Duration (day)	0.048 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.188 ± 0.057 vs 0.16309 (deviation 0.25σ)
Duration fit vs published	0.048 d vs 0.0567 d (deviation 0.19σ)
Mid-time O-C	2.5 min
Depth SNR	1.9
Residual scatter	2.89 %

Light curve

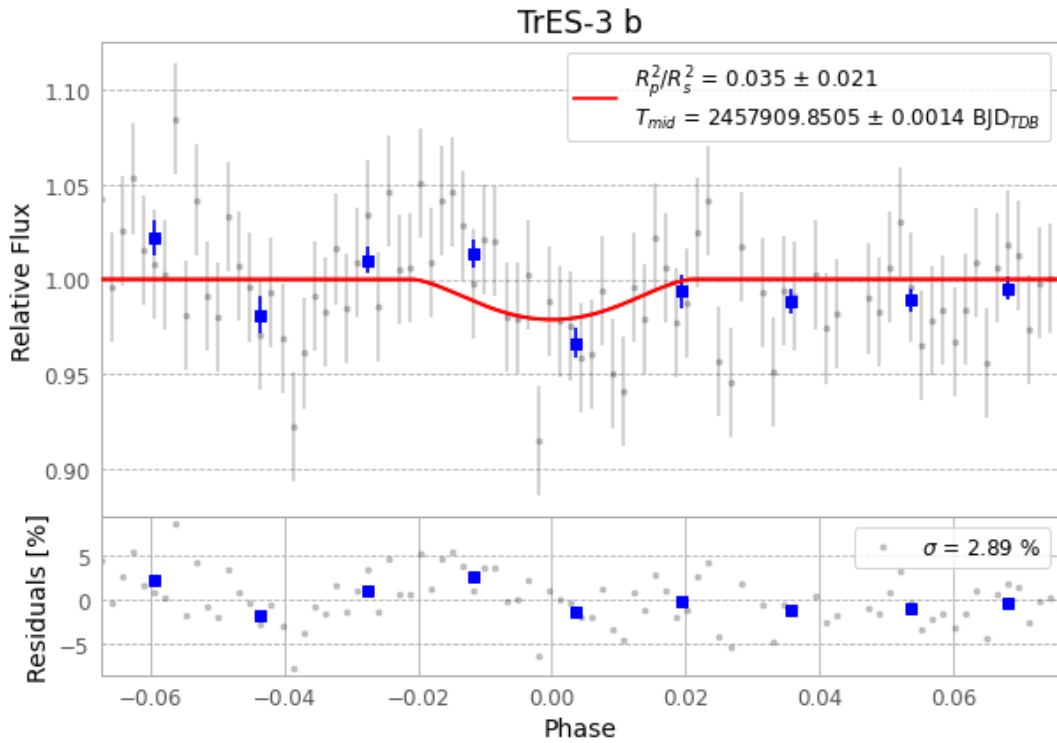


Figure 1 — FinalLightCurve_TrES-3 b_2017-06-04.png (SHA-256 851ff0ba27bc923b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [152, 241], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f71c09d381b5f89c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

91 FITS frames (60 MB), TRES-3170605061212.FITS → TRES-3170605104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170605.log (afd7ad14c64a...), FinalLightCurve_TrES-3 b_2017-06-04.png (851ff0ba27bc...), FinalLightCurve_TrES-3 b_2017-06-04.csv (19f8b9a276c0...)

Compute resources

Wall time 415.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 15:07Z.